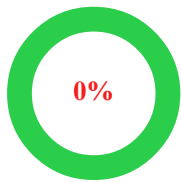


Plagiarism Detection Report by SmallSEOTOOLS



● Plagiarism	0%	● Partial Match	0%
● Exact Match	0%	● Unique	100%

Scan details

Total Words	Total Characters	Plagiarized Sentences	Unique Sentences
577	4946	0	20 (100%)

#1 100% Unique

To develop a Weather Application using JavaScript Fetch API and understand asynchronous programming using Promises, async/await, JSON and try...catch. .

Develop a weather application that accepts a city name from the user, finds its geographical coordinates using a Geocoding API, requests current weather information using the coordinates, and displays the temperature, humidity, wind speed and weather code on the webpage. The application should also handle invalid input, unavailable cities and network/API errors.

Asynchronous programming allows JavaScript to perform time-consuming operations such as API requests without freezing the webpage. An API (Application Programming Interface) provides a way for applications to communicate with external services. In this practical, the fetch() function is used to send network requests and it returns a Promise. The async keyword defines an asynchronous function, while await waits for the Promise to complete. APIs commonly return data in JSON (JavaScript Object Notation) format, which can be converted into JavaScript objects using response.json(). Error handling is performed using try...catch. The received weather data is displayed on the webpage using DOM manipulation and innerHTML

Step for Implementation:

Open VS Code and create a folder named weather-app.

Create three files: index.html, style.css and script.js.

Create an input box for entering the city name.

Create a Search button and weather result container.

Use JavaScript to read the city entered by the user.

Extract latitude and longitude from the API response.

Send another request to the Weather API.

Extract temperature, humidity, wind speed and weather code.

Display the information on the webpage.

Use try...catch to handle errors

Code:

Weather App

Weather App

Search

```
let cityInput = document.getElementById("cityInput");

let searchButton = document.getElementById("searchButton");

let weatherResult = document.getElementById("weatherResult");


searchButton.addEventListener("click", function() {

let city = cityInput.value.trim();


if (city == "") {

weatherResult.innerHTML = "Please enter a city.";

return;

}


getWeather(city);

});


async function getWeather(city) {
```

```
weatherResult.innerHTML = "Loading...";
```

```
try {
```

```
// Geocoding API
```

```
let geoUrl =
```

```
"https://geocoding-api.open-meteo.com/v1/search?name="
```

```
+ encodeURIComponent(city)
```

```
+ "&count=1";
```

```
let geoResponse = await fetch(geoUrl);
```

```
let geoData = await geoResponse.json();
```

```
if (!geoData.results) {
```

```
weatherResult.innerHTML = "City not found.";
```

```
return;
```

```
}
```

```
let location = geoData.results[0].
```

```
let latitude = location.latitude;

let longitude = location.longitude;


// Weather API

let weatherUrl =

"https://api.open-meteo.com/v1/forecast"

+ "?latitude=" + latitude

+ "&longitude=" + longitude

+ "&at=temperature_2m,relative_humidity_2m,"

+ "weather_code,wind_speed_10m";


let weatherResponse = await fetch(weatherUrl);

let weatherData = await weatherResponse.json();


let current = weatherData.current;


let temperature = current.temperature_2m;

let humidity = current.relative_humidity_2m;

let windSpeed = current.wind_speed_10m;

let weatherCode = current.weather_code;
```

```
// Display result

weatherResult.innerHTML =

"" + location.name + "" +

"Country: " + location.country + "" +

"Temperature: " + temperature + " °C" +

"Humidity: " + humidity + " %" +

"Wind Speed: " + windSpeed + " km/h" +

"Weather Code: " + weatherCode + "";

}

catch(error) {

weatherResult.innerHTML =

"Unable to fetch weather data.";

console.log(error);

}

}
```

```
font-family: Arial;  
  
text-align: center;  
  
margin-top: 50px;  
  
}
```

```
input {  
  
padding: 10px;  
  
width: 250px;  
  
}
```

```
button {  
  
padding: 10px 20px;  
  
cursor: pointer;  
  
}
```

```
#weatherResult {  
  
margin-top: 30px;  
  
}
```

Output:

The application successfully accepts a city name and retrieves weather information through external APIs. The Geocoding

API provides latitude and longitude, which are then used by the Weather API to obtain current weather information. The data is converted from JSON into JavaScript objects and displayed dynamically using DOM manipulation.

<https://github.com/juhiparwa-1507/Web-Technology-Lab.git>

Thus, the Weather Application was successfully developed using HTML, CSS and JavaScript. The practical demonstrates asynchronous JavaScript concepts including API, Promise, `fetch()`, `async/await`, JSON, DOM manipulation and `try...catch` error handling.